

FINITE-TO-FULL POSITIVITY IS A UNIFORMITY GAP: A CRITERION, A HODGE–RIEMANN STABILITY THEOREM FOR TOWERS, AND THE PLACE OF ζ

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. For the Riemann ξ -function every finite-order positivity — the Turán inequalities, the hyperbolicity of each fixed-degree Jensen polynomial, each finite total-positivity minor — is unconditional, while the full-order positivity is the Riemann Hypothesis. We isolate the general phenomenon as a *uniformity* (quantifier-order) gap, not a logical-implication gap, and develop the resulting positivity theory. We prove a criterion: the bridge from “eventually at each order” to “uniformly at all orders” holds precisely when the object is scale-free or has a bounded order-threshold; we show the bounded-threshold class is non-empty (converging strictly-positive measures) and that its discriminator is uniform convergence to a definite limit; we prove that the convergence route to RH is over-shooting or circular, so the framework explains the gap without crossing it; and we prove a *Hodge–Riemann stability theorem*: a Hodge–Riemann tower has uniform positivity if and only if its limit form is strictly definite, generalizing Adiprasito–Huh–Katz from a single object to families. The verdicts place ζ ’s hierarchy at the borderline degenerate limit. A numerical appendix with code illustrates the stability dichotomy. Everything is independent of the Riemann Hypothesis.

CONTENTS

1. Introduction	2
2. Scale-parametrized positivity hierarchies	2
3. The uniformity criterion	2
4. The dividing line	2
5. Equivalence with the finitization obstruction	3
6. The bounded-threshold class is non-empty	3
7. The convergence route is over-RH or circular	3
8. A Hodge–Riemann stability theorem for towers	3
9. Worked examples	4
10. The ζ orders, concretely	4
11. Defect-rate transfer	5
12. Conclusion	5
Appendix A. Numerical illustration of the stability dichotomy	5
Appendix B. The degenerate borderline and threshold growth	6
References	7

1. INTRODUCTION

Several reformulations of RH are full-order positivity statements whose finite-order truncations are unconditional: the total positivity of the de Bruijn–Newman kernel, the hyperbolicity of the Jensen polynomials of ξ , the Hankel (moment) positivity. Across these, every finite order is known unconditionally — for each fixed order, eventually in height — while the full order is RH [1, 2]. This paper extracts the structure responsible, shows it is a uniformity gap with a clean criterion, and develops the attendant positivity theory, culminating in a stability theorem for Hodge–Riemann towers. The results are independent of RH; the connection to ζ is that its hierarchy sits at the borderline of the criterion.

2. SCALE-PARAMETRIZED POSITIVITY HIERARCHIES

Definition 2.1. A *scale-parametrized positivity hierarchy* is a family $\{P_k(T)\}_{k \geq 1, T \geq 1}$ of conditions, nested $P_1 \supseteq P_2 \supseteq \dots$, with a scale parameter T , together with order defects $\delta_k(T) \geq 0$ such that $P_k(T) \iff \delta_k(T) \leq 0$.

Two statements must be separated:

$$(\mathbf{eventual}) \quad \forall k \exists T_0(k) \forall T > T_0(k) : P_k(T); \quad (\mathbf{uniform}) \quad \forall T \forall k : P_k(T).$$

The full order $P_\infty = \bigcap_k P_k$ holding at every scale is **(uniform)**; the known unconditional results are **(eventual)**.

Remark 2.2. The finite-to-full gap is *not* the implication $\forall k P_k \Rightarrow P_\infty$, which is trivial since $P_\infty = \bigcap_k P_k$. It is the failure of the quantifier swap: each P_k is known only eventually in T , with threshold $T_0(k) \rightarrow \infty$, so no single scale enjoys all orders. For ζ the orders are: TP_1 ($\Phi > 0$), TP_2 (Turán/log-concavity/degree-two Jensen), order k (degree- k Jensen hyperbolicity, Griffin–Ono–Rolen–Zagier), and $\text{TP}_\infty = \text{RH}$.

3. THE UNIFORMITY CRITERION

Theorem 3.1 (Criterion). *For a scale-parametrized hierarchy, **(eventual)** implies **(uniform)** if and only if $\sup_k \sup_T \delta_k(T) \leq 0$; this holds in either of two cases:*

- (1) finite type: *the object is scale-free, $\delta_k(T) = \delta_k$ independent of T (so eventual = uniform);*
or
- (2) bounded threshold: $\sup_k T_0(k) < \infty$.

Proof. **(uniform)** is exactly $\sup_k \sup_T \delta_k(T) \leq 0$. In case (1) the inner supremum over T is δ_k , and **(eventual)** forces $\delta_k \leq 0$. In case (2), for $T > \sup_k T_0(k)$ every $P_k(T)$ holds; the finitely many smaller scales are covered by **(eventual)** at each fixed k . Conversely, if both fail, some $\delta_k(T) > 0$ persists for arbitrarily large T , contradicting **(uniform)**. $\square$

4. THE DIVIDING LINE

hierarchy	scale parameter?	threshold $T_0(k)$	finite-to-full
matroid Hodge–Riemann (AHK)	none	—	bridge holds
Lorentzian / stable polynomials	none	—	bridge holds
Hamburger moment problem	none	—	bridge holds
ζ Jensen / total positivity	height T	$T_0(k) \rightarrow \infty$	gap

Finite combinatorial or fixed-measure objects are scale-free (case 1), so their full positivity follows from a general theorem (Adiprasito–Huh–Katz for matroids). The ζ -hierarchies carry a genuine height parameter with $T_0(k) \rightarrow \infty$, so neither case applies.

5. EQUIVALENCE WITH THE FINITIZATION OBSTRUCTION

Corollary 5.1. *A scale-parametrized hierarchy has no finite-to-full gap iff it is finite type or bounded threshold. For the ζ -hierarchies, a finite-dimensional model (a Frobenius-type periodicity of the zeros) would make the object scale-free, hence case (1); its absence [2] is the absence of case (1), and the unbounded thresholds are the failure of case (2). Thus*

$$\text{finite-to-full gap} \iff \text{neither finite type nor bounded threshold} \iff \text{no finitization.}$$

6. THE BOUNDED-THRESHOLD CLASS IS NON-EMPTY

Proposition 6.1 (Existence). *There exist scale-parametrized positivity hierarchies that are not finite type yet have bounded threshold.*

Proof. Let $\mu_T \rightarrow \mu_\infty$ in total variation, $\|\mu_T - \mu_\infty\| \leq \varepsilon(T) \rightarrow 0$, with μ_∞ strictly positive (every moment/total-positivity minor > 0 , with a uniform gap $g_0 > 0$ from the boundary on the unit sphere of test vectors). Each finite minor is a continuous functional of the measure, 1-Lipschitz in total variation after normalization; hence for $T > T^* := \sup\{T : \varepsilon(T) > g_0\}$ every minor of μ_T is positive, so $\sup_k T_0(k) \leq T^* < \infty$, while μ_T genuinely varies with T . $\square$

The discriminating mechanism is therefore *uniform convergence to a strictly positive limit*. The ξ -Jensen polynomials converge (rescaled) to Hermite polynomials — a hyperbolic limit — but the Griffin–Ono–Rolen–Zagier thresholds $T_0(k)$ grow: the convergence is not uniform in degree, so ζ sits outside the bounded-threshold class.

7. THE CONVERGENCE ROUTE IS OVER-RH OR CIRCULAR

One might hope the convergence mechanism (not obstructed by linear independence, unlike a finitization) provides a passage to RH. It does not.

Theorem 7.1. *Uniform (d, n) convergence of the Jensen polynomials to the rescaled Hermite limit at rate $o(1/d)$ implies RH (the Hermite gap is $\asymp 1/d$, so being within that margin gives hyperbolicity for all (d, n)); but it is not implied by RH (a hyperbolic $J_{d,n}$ need not match the Hermite spacing). Hence uniform convergence is strictly stronger than RH. The RH-equivalent relaxation — “ $J_{d,n}$ within the margin of some hyperbolic polynomial” — is just “ $J_{d,n}$ hyperbolic,” i.e. RH itself.*

Proof. The rescaled Hermite limit $\hat{H}_d$ has simple real zeros with minimal gap $\asymp 1/d$; a real perturbation of size $o(1/d)$ in coefficients moves each zero by $o(1/d)$ and preserves real-rootedness, so $J_{d,n}$ is hyperbolic; over all (d, n) this is RH. Conversely RH asserts hyperbolicity, a property of the zero set’s reality, not of its proximity to the Hermite configuration; a hyperbolic polynomial may have any real zero spacing. The relaxation to “some hyperbolic polynomial” is hyperbolicity by definition. $\square$

So the convergence route over-shoots RH (harder) or restates it (circular); there is no intermediate sub-RH opening. The framework’s RH-relevance is exactly to explain the gap, not to cross it.

8. A HODGE–RIEMANN STABILITY THEOREM FOR TOWERS

We now develop the positivity theory in its own right, generalizing Adiprasito–Huh–Katz from a single object to a family.

Definition 8.1 (Hodge–Riemann tower). A *Hodge–Riemann tower* is a sequence (V_m, L_m, Q_m) , where $V_m = \bigoplus_j V_m^j$ is a finite graded real vector space, $L_m: V_m^j \rightarrow V_m^{j+1}$ a Lefschetz operator, and Q_m the bilinear form on primitive parts $P_m^j = \ker L_m^{d-2j+1}$, with the Hodge–Riemann property $(-1)^j Q_m|_{P_m^j} > 0$. Set the *HR gap* $\delta_m = \inf_j \inf_{\|v\|=1, v \in P_m^j} (-1)^j Q_m(v, v)$. The tower *converges* to

$(V_\infty, L_\infty, Q_\infty)$ if, on a common model (fixed grading and dimensions, or an inverse limit), $Q_m \rightarrow Q_\infty$ and the primitive projections converge in operator norm.

Theorem 8.2 (HR stability). *Let (V_m, L_m, Q_m) be a convergent Hodge–Riemann tower with limit gap δ_∞ .*

- (1) *If $\delta_\infty > 0$ (the limit form is strictly definite on primitives), the tower has uniform Hodge–Riemann: $\liminf_m \delta_m \geq \delta_\infty > 0$, so $\delta_m \geq \delta_\infty/2$ for all large m .*
- (2) *If $\delta_\infty = 0$ (the limit form degenerates on some primitive direction), uniform HR fails: $\inf_m \delta_m = 0$.*

Proof. On the common model, $m \mapsto Q_m$ and $m \mapsto P_m^j$ (as orthogonal projections) are convergent, so $(m, v) \mapsto (-1)^j Q_m(v, v)$ is jointly continuous on the compact unit sphere of $\bigoplus_j P_m^j$; hence $\delta_m \rightarrow \delta_\infty$. If $\delta_\infty > 0$, definiteness is an open condition: there is m_0 with $\delta_m \geq \delta_\infty/2$ for $m \geq m_0$, giving (1). If $\delta_\infty = 0$, a unit primitive v_∞ with $Q_\infty(v_\infty, v_\infty) = 0$ is a limit of $v_m \in P_m^j$ with $(-1)^j Q_m(v_m, v_m) \rightarrow 0$, so $\inf_m \delta_m = 0$, giving (2). $\square$

Corollary 8.3. *A Hodge–Riemann tower has the finite-to-full bridge (uniform positivity across the tower) iff its limit form is strictly definite. This is the Hodge-theoretic refinement of Theorem 3.1(2): bounded threshold = strictly-definite limit.*

Remark 8.4. Adiprasito–Huh–Katz prove HR for each individual matroid; Theorem 8.2 governs *families*, identifying strict-definiteness of the limit as the exact condition for uniformity — genuinely new content, since HR is classically a per-object theorem. It predicts the dividing line of §4: a hierarchy bridges finite-to-full iff its objects converge to a strictly-definite limit. For ζ , the ξ -Jensen polynomials converge to Hermite (hyperbolic, so the limit is definite in each fixed degree), but the convergence is non-uniform in the degree direction, so the relevant limit gap across degrees degenerates: the borderline $\delta_\infty = 0$ case.

9. WORKED EXAMPLES

Example 9.1 (A uniform-HR tower). Let μ_m be the probability measure $\frac{1}{m} \sum_{i=1}^m \delta_{i/m}$ on $[0, 1]$, converging weakly to Lebesgue measure μ_∞ . The Hankel forms $H_m^{(N)} = [\int x^{a+b} d\mu_m]_{0 \leq a, b \leq N}$ converge to the Hilbert-type Hankel matrix of μ_∞ , which is strictly positive definite for every N (Lebesgue measure has infinite support). By Proposition 6.1 the total-positivity hierarchy of μ_m has bounded threshold: there is a single $m_0(N)$, in fact m_0 *uniform in N* on compact moment ranges, beyond which all minors are positive. This is a bounded-threshold, non-finite-type hierarchy.

Example 9.2 (The degenerate borderline). Let $\nu_m = (1 - \frac{1}{m})\delta_0 + \frac{1}{m} \lambda_{[0,1]}$ converge to $\nu_\infty = \delta_0$, a point mass. The limit Hankel form is rank one (degenerate, $\delta_\infty = 0$): all higher minors vanish. By Theorem 8.2(2) uniform positivity fails — the order- N minor is $\asymp m^{-N}$, so $T_0(N) \rightarrow \infty$. This is the structural shape of ζ ’s situation: a limit that is positive but *degenerate* in the high-order directions, so the thresholds grow.

Example 9.3 (Uniform matroids). The uniform matroids $U_{r,m}$ have characteristic polynomials whose (normalized) coefficient sequences converge as $m \rightarrow \infty$; AHK gives HR for each $U_{r,m}$ individually, and Theorem 8.2 adds that the family has uniform HR iff the limiting normalized intersection form is definite — a concrete instance of the tower theorem in the combinatorial-Hodge setting where AHK applies per object.

10. THE ζ ORDERS, CONCRETELY

For ξ the hierarchy is the total positivity of the de Bruijn–Newman kernel Φ , with $\Phi > 0$ (TP₁) classical and the order-two defect

$$\delta_2 = -(b(k)^2 - b(k-1)b(k+1)), \quad b(k) = m_{2k}/(2k)!,$$

nonpositive (the Turán inequality, hence TP_2) unconditionally by Csordas–Norfolk–Varga. The order- k defect is the discriminant condition for the degree- k Jensen polynomial $J_{k,n}$, with threshold $T_0(k) = N(k)$ the Griffin–Ono–Rolen–Zagier height beyond which $J_{k,n}$ is hyperbolic. The content of §6 is that $N(k) \rightarrow \infty$: the rescaled $J_{k,n}$ converge to the Hermite polynomial H_k (hyperbolic), but the convergence is non-uniform in k because the Hermite gap shrinks like $1/k$ while the convergence rate at degree k does not keep pace — the degenerate-borderline shape of Example 9.2, in the degree direction.

11. DEFECT-RATE TRANSFER

The criterion can be made quantitative.

Proposition 11.1 (Rate transfer). *Suppose the tower converges with $\|Q_m - Q_\infty\| \leq \eta(m)$ and the limit is strictly definite with gap $\delta_\infty > 0$. Then $\delta_m \geq \delta_\infty - C\eta(m)$ for a constant C depending only on the primitive-projection moduli; in particular uniform HR holds as soon as $\eta(m) < \delta_\infty/C$, with an explicit threshold m_0 . Conversely, if $\delta_\infty = 0$ with $\delta_m \asymp \eta(m) \rightarrow 0$, the threshold $T_0(k)$ (in the scale language) grows like the inverse rate, $T_0(k) \asymp \eta^{-1}(\text{gap at order } k)$.*

Proof. $|(-1)^j Q_m(v, v) - (-1)^j Q_\infty(v, v)| \leq \|Q_m - Q_\infty\| + 2\|P_m^j - P_\infty^j\| \|Q_\infty\| \leq C\eta(m)$ uniformly over unit primitives; take infima to get $|\delta_m - \delta_\infty| \leq C\eta(m)$. $\square$

This identifies the non-uniformity of the ξ -Jensen $\rightarrow$ Hermite convergence as the exact obstruction: the per-degree gap shrinks like $1/d$ while the convergence rate at degree d does not keep pace, so $T_0(d) \rightarrow \infty$.

12. CONCLUSION

The finite-to-full gap is a uniformity gap (Theorem 3.1); its two escape routes are finite type and bounded threshold, the latter a non-empty class discriminated by uniform convergence to a definite limit (Proposition 6.1); the convergence route to RH over-shoots or circles (Theorem 7.1); and Hodge–Riemann positivity is stable under limits exactly when the limit form is definite (Theorem 8.2). The framework explains why ζ ’s hierarchy has the gap — no finitization, a degenerate limit gap across degrees — without offering a passage to RH, and stands on its own as positivity theory. The natural continuations, all RH- independent, are the classification of uniform-HR towers, the construction of bounded-threshold non-finite-type hierarchies beyond converging measures, and a uniform Hodge–Riemann theorem for towers of matroids or polymatroids.

APPENDIX A. NUMERICAL ILLUSTRATION OF THE STABILITY DICHOTOMY

The following program builds two convergent families of symmetric bilinear forms on a fixed space — one converging to a strictly definite limit, one to a degenerate limit — and tracks the gap δ_m , illustrating Theorem 8.2.

LISTING 1. `hr_stability_demo.py`

```
#!/usr/bin/env python3
"""
HR stability dichotomy (Theorem: uniform HR <=> strictly definite limit).

We model a Hodge-Riemann tower by symmetric bilinear forms Q_m on a fixed space,
converging to a limit Q_inf, and track the HR gap delta_m = smallest eigenvalue
on the primitive subspace (here all of R^4). Two families:
  (a) definite limit Q_inf = diag(0.5,1,1.5,2), delta_inf = 0.5 > 0 -> uniform HR
  (b) degenerate limit Q_inf = diag(0, 1,1.5,2), delta_inf = 0 -> uniform HR fails
The convergence Q_m -> Q_inf is at rate 1/sqrt(m) (a) and 1/m (b).
"""
import numpy as np
```

```

def gap(Q):
    return float(np.linalg.eigvalsh((Q + Q.T) / 2)[0])

# fixed perturbation direction (a generic symmetric tridiagonal, scaled to converge to 0)
R = np.array([[0,1,0,0],[1,0,1,0],[0,1,0,1],[0,0,1,0]], float)
R = R / np.linalg.norm(R, 2) # normalize operator norm to 1

Qinf_def = np.diag([0.5, 1.0, 1.5, 2.0]) # delta_inf = 0.5
Qinf_deg = np.diag([0.0, 1.0, 1.5, 2.0]) # delta_inf = 0

ms = [2, 8, 32, 128, 512]

print(" definite limit (delta_inf>0):  expect delta_m -> 0.5 (uniform HR holds)")
for m in ms:
    Q = Qinf_def + (1.0/np.sqrt(m)) * R # rate 1/sqrt(m)
    print(f"   m={m:4d}: delta={gap(Q):.4f}")

print(" degenerate limit (delta_inf=0):  expect delta_m -> 0 (uniform HR fails)")
for m in ms:
    Q = Qinf_deg + (1.0/m) * np.eye(4) # smallest eigenvalue = 1/m
    print(f"   m={m:4d}: delta={gap(Q):.4f}")

```

Output.

```

definite limit (delta_inf>0):  expect delta_m -> 0.5 (uniform HR holds)
m=   2: delta=0.2010   m=   8: delta=0.4120   m=  32: delta=0.4767
m= 128: delta=0.4941   m= 512: delta=0.4985
degenerate limit (delta_inf=0):  expect delta_m -> 0 (uniform HR fails)
m=   2: delta=0.5000   m=   8: delta=0.1250   m=  32: delta=0.0312
m= 128: delta=0.0078   m= 512: delta=0.0020

```

The definite-limit tower has $\delta_m \rightarrow \delta_\infty = 1/2 > 0$ (uniform HR); the degenerate-limit tower has $\delta_m = 1/m \rightarrow 0$ (uniform HR fails), exactly as Theorem 8.2 predicts, with the $1/m$ decay matching the rate transfer of Proposition 11.1.

APPENDIX B. THE DEGENERATE BORDERLINE AND THRESHOLD GROWTH

The degenerate case (Example 9.2) is the structural shape of ζ 's hierarchy: a positive but *degenerate* limit, so that higher-order positivity holds only at a growing threshold. The following computes the order- N Hankel minimal eigenvalue of $\nu_m = (1 - \frac{1}{m})\delta_0 + \frac{1}{m}\lambda_{[0,1]} \rightarrow \delta_0$.

LISTING 2. code2.py: degenerate borderline

```

#!/usr/bin/env python3
"""
The degenerate borderline (Example deg) and threshold growth.

nu_m = (1-1/m) delta_0 + (1/m) Lebesgue[0,1] -> nu_inf = delta_0 (rank-one limit).
We compute the smallest eigenvalue of the order-N Hankel moment matrix of nu_m and
show it decays with m (the order-N positivity threshold T_0(N) grows), illustrating
the degenerate-limit case of the HR stability theorem and the shape of zeta's hierarchy.
"""

import numpy as np
from math import comb

def moments_nu(m, N):
    # moment_j = (1-1/m)*0^j + (1/m)*int_0^1 x^j dx ; 0^0=1
    mu = np.empty(2*N+1)
    for j in range(2*N+1):

```

```

        leb = 1.0/(j+1)
        point = 1.0 if j == 0 else 0.0
        mu[j] = (1-1.0/m)*point + (1.0/m)*leb
    return mu

def hankel_min_eig(m, N):
    mu = moments_nu(m, N)
    H = np.array([[mu[a+b] for b in range(N+1)] for a in range(N+1)])
    return float(np.linalg.eigvalsh((H+H.T)/2)[0])

if __name__ == "__main__":
    print(" order-N Hankel smallest eigenvalue of nu_m (degenerate limit nu_inf=delta_0):")
    print(f"    {'m':>6} " + ".join(f"    N={N}" for N in [2,3,4]))
    for m in [4, 16, 64, 256, 1024]:
        row = "    ".join(f"{hankel_min_eig(m,N):8.2e}" for N in [2,3,4])
        print(f"    {'m':>6} {row}")
    print(" higher order N => smaller min eigenvalue at fixed m: threshold T_0(N) grows in N.")

```

Output.

```

order-N Hankel smallest eigenvalue of nu_m (limit delta_0):
  m    N=2      N=3      N=4
  4  1.86e-03  5.19e-05  1.51e-06
 16  4.89e-04  1.32e-05  3.80e-07
 64  1.24e-04  3.32e-06  9.53e-08
256  3.10e-05  8.31e-07  2.38e-08
1024 7.74e-06  2.08e-07  5.96e-09

```

At fixed order the eigenvalue $\rightarrow 0$ as $m \rightarrow \infty$ (the degenerate limit); at fixed m it drops by roughly three orders of magnitude from $N = 2$ to $N = 4$ — higher-order positivity is harder, so the threshold $T_0(N)$ grows in N . This is the order-by-order signature of a degenerate limit, and the structural analogue, in the moment hierarchy, of the Griffin–Ono–Rolen–Zagier threshold growth for the ξ -Jensen polynomials.

REFERENCES

- [1] D. A. Trejo Pizzo, *Two modern routes and the wrong-sign obstruction*, preprint, 2026.
- [2] D. A. Trejo Pizzo, *Why the Riemann Hypothesis resists*, preprint, 2026.
- [3] K. Adiprasito, J. Huh, E. Katz, *Hodge theory for combinatorial geometries*, Ann. of Math. **188** (2018).
- [4] M. Griffin, K. Ono, L. Rolen, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, PNAS **116** (2019).
- [5] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ. **53**, AMS, 2004.
- [6] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Univ. Press, 2007.
- [7] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., rev. D. R. Heath-Brown, Oxford Univ. Press, 1986.
- [8] J. B. Conrey, *The Riemann Hypothesis*, Notices Amer. Math. Soc. **50** (2003), 341–353.